

OS A01 The National Information Ecosystem

Canonical learner lesson

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Advanced professional level for statisticians methodologists data stewards and domain specialists

This lesson develops the ability to map a national information ecosystem and to use AI responsibly to accelerate evidence extraction, actor and asset classification, flow analysis, risk detection and documentation. AI remains assistive. The statistician and relevant subject-matter specialists retain authority for concepts, methods, evidence, validation, interpretation and every consequential decision.

Controlling references: Phase 2A Curriculum Blueprint and Competency Architecture; SRAI Official Statistics and AI Pathway Production Reference version 1.3; Pathway-Wide AI-Assisted Statistical Work Standard version 1.0.

Module purpose and professional outcome

Official statistics operate within a national information ecosystem rather than inside one institution. The ecosystem includes public producers, administrative source owners, regulators, research bodies, private data holders, infrastructure providers, funders, users and affected communities. Their mandates, incentives, technical systems and relationships determine whether information is timely, coherent, interpretable and fit for public decisions.

The professional outcome is a scoped ecosystem map with stable identifiers, authoritative evidence, critical information flows, user needs, quality and governance risks, coordination gaps and a justified statement of where AI may or may not add value. The map is not a decorative stakeholder diagram. It is an analytical product that supports institutional decisions about data access, standards, investment, quality improvement, pilot selection and accountability.

Learners will perform both a conventional workflow and an AI-assisted workflow. They will compare quality-adjusted time, record prompts and corrections, and demonstrate that the final judgement remains human.

Field	Requirement
Entry level	Basic familiarity with a statistical organization or sector; no prior AI expertise required.
Exit level	Guided practitioner able to complete a defensible ecosystem map and supervised AI-assisted analysis.
Structured time	Approximately 14 hours of study and facilitated practice plus a workplace assignment.
Primary product	Scoped ecosystem map, actor and asset registers, flow analysis, fitness matrix and AI opportunity decision.
Assessment	Professional product, evidence trace, prompt record, verification log and oral defence.

Learning outcomes and required evidence

Competence is demonstrated through an integrated product. Correct terminology is necessary but insufficient. The learner must show how a public information need becomes a statistical requirement, how institutions and assets contribute to it, how errors and delays propagate, and how AI assistance is bounded by evidence and professional authority.

Each outcome is assessed through observable evidence. Unknowns are preserved rather than silently completed. Conflicting sources are represented as conflicts. An AI-generated suggestion receives no evidentiary weight until it is checked against an authoritative source or reproducible test.

Learner can	Required evidence
Define an ecosystem boundary	Scope statement naming decision, domain, geography, period, users and exclusions.
Map actors and authority	Actor register with roles, mandates, evidence labels and unresolved authority questions.
Inventory information assets	Asset register with provenance, population, identifiers, transformations, quality and access conditions.
Trace critical flows	Flow record showing handoffs, controls, delays, failure modes, responsibility and fallback.
Translate user needs	Fitness-for-use matrix with quality requirements and consequences of failure.
Use AI responsibly	Prompt record, candidate outputs, verification, corrections, limits and non-use decisions.
Defend a recommendation	Prioritized improvement and bounded AI opportunity decision signed off by accountable specialists.

How to use this lesson and safeguards

Work through the conceptual chapters before opening the practical laboratory. The lesson deliberately separates claims that are evidenced, reported and unknown. This discipline is essential because ecosystem mapping commonly relies on laws, administrative documentation, interviews, system records and professional knowledge of unequal reliability.

Use only approved AI tools and environments. Public or synthetic information may be used in the Kintaba laboratory. Real institutional work requires data classification before any material is supplied to an external system. When source documents contain personal data, confidential statistical information, security details or restricted operational information, use an approved secure environment or exclude the material.

Prompting is treated as specification work. A good prompt does not transfer authority to the model. It states the task, evidence boundary, output schema, uncertainty rules and prohibited inferences. Every output remains a candidate until the responsible statistician and subject specialist have verified and accepted it.

- Never place confidential records, passwords, personal data or security configurations in an unapproved AI tool.
- Do not convert informed opinion into fact. Preserve evidenced, reported and unknown labels.
- Map actual decision authority; do not assume the national statistical office controls every source.
- Do not credit time saved by bypassing review, weakening validation or hiding uncertainty.

The national information ecosystem as an analytical object

A national information ecosystem is the connected set of institutions, people, rules, data assets, technologies, processes, funding arrangements and relationships through which information is generated, transformed, governed, disseminated and used. Official statistics are a central component, but the ecosystem is wider than the national statistical office and wider than the legally designated national statistical system.

The ecosystem perspective changes the unit of analysis. A conventional organizational review asks whether a particular agency has sufficient staff, systems and procedures. Ecosystem analysis asks whether the complete information chain can satisfy a public need. A technically strong producer may still fail users when source owners change definitions without notice, identifiers cannot be linked, legal access is unresolved or a critical platform has no sustainable operator.

The map therefore represents both structure and performance. Structure identifies actors, assets, mandates and connections. Performance examines coverage, error, timeliness, coherence, accessibility, resilience and actual use. Institutional authority and evidentiary strength must be encoded, not inferred from visual prominence.

Analytical layer	Central question
Public value	Which decisions, rights, services or debates require information?
Production	Which assets and transformations produce the statistical product?
Governance	Who may decide, access, change, approve, release and challenge?
Capability	Which skills, systems, funding and suppliers sustain the process?
Risk	Where can failure, bias, delay, misuse or discontinuity occur?

Systems thinking feedback and emergence

Information ecosystems exhibit interdependence, feedback and emergent behaviour. A delay in an administrative source can change the sample frame, which changes coverage, which affects estimates, which changes user trust and may reduce future reporting cooperation. These effects cannot be understood by listing institutions independently.

Feedback can stabilize or destabilize the system. Revision policies, metadata feedback and user consultation may improve quality through learning. Political pressure, fragmented funding and parallel donor systems may reward speed or visibility at the expense of comparability and maintenance. Analysts should identify incentives and feedback rather than treating every gap as a technical deficiency.

Emergence matters for AI. An apparently accurate classification model may alter staff behaviour, create automation bias or shift resources away from hard-to-classify cases. Ecosystem assessment therefore considers second-order consequences: who adapts, which records change, what evidence disappears and who becomes dependent on a supplier or model.

System property	Mapping implication
Interdependence	Trace dependencies and error propagation across institutional boundaries.
Feedback	Record how users, revisions, audits and incentives change production behaviour.
Adaptation	Expect actors and data-generating processes to respond to new rules and tools.
Emergence	Assess consequences not visible from component-level performance.
Path dependence	Document legacy classifications, systems and agreements that constrain change.

Boundary definition and nested views

A useful boundary is justified by the decision and analytical purpose. Boundaries that are too narrow hide critical source owners, users and risks. Boundaries that are too broad create inventories without decision value. State the focal decision, statistical product, population, geography, reference period, institutional scope and exclusions before collecting records.

Three nested views are useful. The national view shows cross-government producers, standards and major flows. The domain view examines a sector such as food and agriculture while retaining links to finance, geospatial infrastructure and civil registration. The product or use-case view follows one decision chain, such as a seasonal crop forecast, from observation to public action.

Boundaries remain provisional during discovery. New evidence may justify adding an actor or excluding a peripheral source. Changes must be recorded with reasons so that apparent differences between versions are not mistaken for substantive institutional change.

View	Boundary	Typical use
National	Major producers, sources, infrastructures and users across domains.	Coordination, legislation, standards and investment planning.
Domain	One sector and its interfaces with the wider system.	Sector strategy and cross-source integration.
Product or use case	One statistical output or decision chain.	Operational improvement and pilot design.

Actors institutions and roles

Institutions are not roles. One institution may own a source, produce a statistic, operate a platform and use the result. Conversely, methodological authority may be distributed across legislation, professional standards and interagency agreements. Mapping only organization names conceals accountability conflicts.

Assign roles separately and permit multiple roles per actor. For each assignment, record the legal or operational basis, the decision rights involved and whether the evidence is current. A ministry may be the operational owner of an administrative register while the national statistical office defines statistical transformations. A technology supplier may operate infrastructure without authority to approve concepts or outputs.

Role analysis must also identify affected groups. Farmers, firms, households and local administrations generate information and experience consequences when classifications, eligibility rules or risk flags are wrong. Their position cannot be reduced to passive data provision.

Role	Accountability question
Source owner	Who controls the original operational or observed data?
Producer	Who transforms sources into a statistical output?
Methodological authority	Who approves concepts, methods, revisions and release?
Steward	Who maintains metadata, access, quality and change communication?
Coordinator	Who sets shared standards and resolves duplication?
Technology provider	Who operates platforms, models or infrastructure?
User or affected group	Who decides with the information or bears consequences?

Authority governance and institutional accountability

Authority should be decomposed into specific decision rights: define concepts, compel or authorize collection, access records, transform data, approve methods, release results, revise figures and suspend a process. Formal mandates may differ from operational practice. Both must be recorded because a lawful design can still fail when implementation authority is unclear.

Governance quality depends on escalation and challenge. The map should identify who can question a method, require evidence, reject a release, investigate a breach or resolve disagreement between producers. Absence of an escalation route is a material risk even when routine production appears stable.

AI does not acquire institutional authority by generating an answer. It may summarize mandates or highlight apparent inconsistencies, but qualified officials must interpret the law, confirm current delegations and decide how authority applies. Legal and policy questions require authorized advice.

Decision right	Evidence
Define	Law, standard, approved methodology or classification governance record.
Access	Statutory authority, agreement, consent basis or approved protocol.
Transform	Documented process ownership and reproducible procedure.
Approve	Named accountable office, committee or delegated professional role.
Release and revise	Publication policy, revision calendar and authorization record.
Challenge and escalate	Audit, appeal, incident or dispute-resolution mechanism.

Evidence architecture and uncertainty labels

Ecosystem maps combine heterogeneous evidence. Legislation establishes formal mandates but may not show current practice. Interviews reveal operational realities but may be incomplete or strategically framed. System logs show events but not always their meaning. A defensible map preserves these distinctions.

Use stable evidence identifiers and retain sufficient provenance for another reviewer to locate the source. Record title, owner, date, version, location, access condition and the claim supported. Evidence should support a precise statement rather than merely appear in a bibliography.

Conflicts are analytical results. When two agencies report different production estimates, do not average or select one silently. Record concept, population, reference period, method, revision status and authority. The unresolved discrepancy may be more decision-relevant than either number.

Label	Meaning	Minimum record
E Evidenced	Supported by an authoritative document, dataset, system record or approved observation.	Source ID and exact location.
R Reported	Stated by a knowledgeable respondent but not independently confirmed.	Role, date and interview record.
U Unknown	Unavailable, conflicting or outside current scope.	Question, consequence and proposed evidence owner.
I Inferred	Analytical conclusion derived from stated evidence.	Reasoning chain and reviewer.

Information assets and statistical provenance

An information asset is a governed body of data, metadata, code, documentation or output that contributes to an information need. The producing institution is not the asset, and a publication is not a substitute for its underlying sources, transformations and quality evidence.

An asset record should support provenance and fitness decisions. Stable identifiers permit reuse across actor, flow, risk and user-need records. Versioning is essential because coverage, concepts, identifiers and collection processes change. The record should show when a fitness conclusion was made and for which use.

Reference assets deserve equal attention. Classifications, frames, geographies, code lists, concordances, metadata and quality reports determine whether sources can be integrated. They often become invisible infrastructure until a change breaks comparability.

Required field	Purpose
Identity and owner	Establishes the governed object and accountable steward.
Population unit coverage	Defines who or what is represented and excluded.
Time geography frequency	Supports alignment, timeliness and comparability.
Concepts classifications identifiers	Controls meaning and integration.
Transformations and versions	Supports reproducibility and change analysis.
Quality access confidentiality	Determines permissible and defensible use.
Dependencies and users	Reveals operational and decision consequences.

Surveys censuses and designed data

Designed statistical collections use explicit populations, frames, instruments and estimation procedures. Their strength is not automatic accuracy but transparent design. Ecosystem mapping must connect collection design to coverage, measurement, nonresponse, processing, estimation and revision risks.

The frame is an ecosystem dependency. It may draw on an earlier census, administrative registers, geospatial enumeration areas or commercial listings. Frame deterioration affects inclusion probabilities and may create persistent undercoverage. AI-assisted matching can suggest duplicates or updates, but the statistician must define matching tolerances, review uncertain cases and assess induced coverage bias.

Instrument design and fieldwork also generate metadata. Paradata, interviewer observations, call records and edit histories can explain error patterns. They require governance because they may contain personal or operationally sensitive information.

Risk family	Ecosystem question
Coverage	Which units can enter the frame, and which groups are systematically missed?
Measurement	How do concepts, wording, devices and respondents affect observed values?
Nonresponse	Who fails to respond and how does adjustment depend on auxiliary information?
Processing	Which edits, coding and transformations can introduce error?
Estimation	Which assumptions support weights, models and variance estimates?
Revision	How are late data and method changes communicated to users?

Administrative records and operational processes

Administrative records are generated to deliver services, enforce rules, collect revenue or manage programmes. Statistical use is secondary. Their populations, concepts and update patterns therefore follow operational incentives and legal eligibility rather than statistical definitions.

Quality assessment begins with process knowledge. Identify the purpose, legal basis, transaction points, responsible offices, identifiers, validation rules, backlogs, appeals and changes in policy or systems. A stable record count can conceal changing eligibility or delayed entry. Conversely, an operational discontinuity may be informative rather than an error.

AI can accelerate metadata extraction, compare field definitions and classify change notices. It should not infer population coverage or conceptual equivalence from variable names alone. The administrative owner explains operations; the statistician evaluates fitness for the intended statistical use.

Assessment area	Evidence sought
Purpose and population	Eligibility, service rules and units actually represented.
Concepts and identifiers	Definitions, codes, uniqueness and linkage behaviour.
Process controls	Entry, validation, correction, appeals and backlog management.
Change	Policy, software, staffing and organizational changes.
Quality signals	Missingness, duplicates, lag, discontinuities and reconciliation records.
Access and continuity	Legal access, extraction stability, stewardship and fallback.

Geospatial sensor private and research sources

Alternative sources can improve spatial or temporal resolution but introduce distinct selection, measurement and sustainability risks. Satellite classifications are model outputs rather than direct observations. Sensor networks depend on placement, calibration and uptime. Private transactions represent customers and business processes, not automatically the target population.

Research results require external-validity analysis. Performance measured on accessible districts, research stations or labelled cases may not generalize to remote areas, rare crops or different seasons. Record the sampling mechanism, validation domain and conditions under which extrapolation is proposed.

Contracts and funding are quality variables. A technically useful private or research source may be unsuitable for recurrent official statistics if access, methods, skills or infrastructure cannot be sustained. AI may help compare documentation and generate test cases, while specialists determine representativeness and acceptable generalization.

Source	Critical question
Geospatial model	What reference labels, domains, resolution and error measures support the result?
Sensor	Where is it located, how is it calibrated and how are outages handled?
Private data	Which population and behaviours enter the system, and under what incentives?
Research evidence	What design supports causal or descriptive claims, and where does it generalize?
Experimental source	What is required to move from exploration to controlled statistical production?

Integrated sources linkage and uncertainty propagation

Integration may combine surveys, administrative records, geospatial sources and models. It creates value by improving coverage or timeliness, but errors can compound. Linkage errors, duplicates, temporal misalignment and classification differences must be represented explicitly.

Separate deterministic linkage, probabilistic linkage and model-based integration. For each method, document identifiers, blocking rules, thresholds, clerical review, missed-match and false-match evidence, and the consequences of error. High aggregate match rates do not guarantee unbiased results for small domains.

Uncertainty from component sources should propagate to the final product. When formal propagation is infeasible, structured sensitivity analysis and limitation statements are required. AI-generated code can accelerate comparisons and diagnostics, but statisticians retain authority for the estimand, linkage design, uncertainty treatment and release.

Integration control	Required record
Alignment	Population, unit, geography, concept and reference-time compatibility.
Linkage	Identifiers, rules, thresholds, review and error evidence.
Duplication	Entity resolution and treatment of multiple records.
Transformation	Classification concordances, derivations and version control.
Uncertainty	Sampling, linkage, model and measurement components.
Disposition	Accept, restrict, revise, investigate or reject for the stated use.

Information flows lineage and handoffs

A flow connects an initiating event to a decision or release through sources, transformations, controls and handoffs. The purpose is to reveal where meaning, quality, time or authority changes. A flow diagram without identifiers and evidence is illustrative, not auditable.

At each handoff, record source and destination actors and assets, transformation, control, timing, failure mode, responsibility and fallback. Handoffs across organizations deserve special attention because assumptions may be undocumented and correction authority may be unclear.

Lineage supports incident response and revision. When an input changes, the institution should know which outputs and users are affected. AI may draft lineage from documentation or code, but the output must be reconciled with actual systems and staff knowledge.

Flow element	Record
Trigger	Event or schedule that initiates production.
Source and destination	Actor and asset IDs at each handoff.
Transformation	Editing, coding, linkage, modelling, aggregation or interpretation.
Control	Validation, approval, access or quality check.
Timing	Frequency, deadline, delay and synchronization.
Failure and fallback	Error mode, correction owner and continuity route.

Metadata standards and interoperability

Interoperability is semantic, organizational and technical. Common file formats do not establish common meaning. Systems can exchange records successfully while using different populations, classifications or time references. Ecosystem mapping should distinguish syntactic compatibility from conceptual comparability.

Metadata must be governed across versions. A variable catalogue should link definitions to classifications, derivations, valid periods and responsible owners. Concordances require evidence because one-to-many mappings can change aggregates and time series.

AI can extract candidate metadata from forms, manuals, code and database schemas. It is useful for detecting inconsistent labels and drafting crosswalks. It should not approve equivalence. Statisticians and domain specialists must confirm definitions, permissible transformations and consequences for comparability.

Layer	Requirement
Semantic	Common or explicitly reconciled concepts, units and classifications.
Organizational	Agreed ownership, change communication and issue resolution.
Technical	Stable schemas, identifiers, interfaces and version control.
Statistical	Preserved population, estimand, uncertainty and comparability.
Governance	Authorized access, retention, confidentiality and auditability.

User needs and fitness for use

Quality is relational: information is fit or unfit for a stated use under stated conditions. Begin with the decision, not the available data. Identify the user, action, population, geography, reference period, timing and consequence of error. Translate these into measurable or reviewable requirements.

Different users can legitimately require different products. An early warning may prioritize timeliness and directional accuracy, while compensation or regulatory decisions require unit-level precision, transparency and appeal. The ecosystem map should not collapse these requirements into a single quality score.

AI may summarize consultations and propose a requirements matrix. Human users and specialists confirm the decision context, tolerances and distributional consequences. Marginalized or affected groups may reveal needs missed by institutional users.

Dimension	Requirement question
Relevance	Does the product address the actual decision and population?
Accuracy and reliability	What errors and revisions are tolerable?
Timeliness and punctuality	When must information be available and released?
Coherence and comparability	With which series, geographies and classifications must it align?
Accessibility and interpretability	Can intended users obtain and correctly understand it?
Equity and impact	Which groups may be missed or harmed by error?

Quality frameworks total error and assurance

A quality framework organizes dimensions; a total-error perspective explains mechanisms. Coverage, measurement, nonresponse, processing, linkage and model error interact with revisions and communication. Ecosystem analysis connects these mechanisms to actors and controls.

Assurance should be proportionate to consequence. Low-risk exploratory outputs may tolerate wider uncertainty and restricted circulation. Official releases affecting resources or rights require stronger benchmarks, independent review, reproducibility and disclosure control.

AI-assisted quality work includes drafting tests, detecting anomalies, comparing documentation and generating edge cases. The statistician decides which deviations are errors, whether assumptions hold and whether residual uncertainty is acceptable. Automated checks are evidence, not approval.

Assurance level	Example control
Routine	Schema, range, completeness and reconciliation checks.
Methodological	Design review, benchmark comparison and sensitivity analysis.
Independent	Separate reproduction, challenge review or external benchmark.
Operational	Access, continuity, incident and change-management controls.
Release	Final figures, narrative, confidentiality and authorization sign-off.

Risk resilience and sustainability

Risk combines likelihood, consequence, detectability and capacity to recover. Ecosystem risks include single-person dependencies, expiring donor funding, obsolete classifications, supplier lock-in, weak agreements, cyber incidents and loss of public trust. Prioritization must consider affected decisions and populations.

Resilience is the ability to continue or recover while preserving essential quality and accountability. Record fallback sources, manual procedures, backups, alternate expertise and thresholds for degraded service. A fallback is not credible unless it is tested and resourced.

AI can help enumerate scenarios and organize risk evidence. It may overproduce generic risks or exaggerate unlikely events. Specialists select plausible scenarios, assess consequences and approve mitigations. Sustainability analysis should include recurring cost, local skills, model maintenance, data access and exit options.

Risk domain	Example
Institutional	Unclear authority or coordination failure.
Methodological	Unstable concepts, frames or unvalidated model.
Operational	Late handoff, staff dependency or untested fallback.
Technology	Obsolescence, outage, cybersecurity or vendor lock-in.
Financial	Temporary funding for recurrent production.
Trust	Opaque revisions, inequitable impact or misuse.

Confidentiality security ethics and lawful use

Ecosystem mapping can expose sensitive details even without microdata. System names, vulnerabilities, access routes and data-sharing disputes may require restricted handling. Classify the mapping records and separate public descriptions from controlled operational evidence.

Statistical confidentiality limits disclosure and use; cybersecurity protects systems and information; privacy and data-protection law govern personal data; professional ethics addresses fairness, harm, transparency and public trust. These regimes overlap but are not interchangeable.

Before AI use, determine whether content may leave the institutional environment, whether the provider retains inputs, how access is controlled and whether outputs can be audited. Exclusion is a legitimate design choice. Legal, security and data-stewardship authorities approve conditions; the model cannot interpret itself into compliance.

Decision	Evidence
Use approved tool	Data classification and authorized environment recorded.
Use redacted or synthetic material	Transformation documented and residual disclosure risk reviewed.
Use secure internal capability	Access, logging, retention and model controls approved.
Do not use AI	Risk, legal uncertainty or evidence sensitivity exceeds benefit.
Escalate	Authority or policy is unclear and consequential use is proposed.

AI foundations for ecosystem analysis

AI is a family of capabilities rather than a single method. Rules automate explicit decisions. Machine learning estimates patterns from data. Generative models produce candidate text, code or structure. Retrieval systems ground responses in supplied collections. Agentic workflows sequence tools and actions. Each has different evidence, control and failure requirements.

For ecosystem mapping, generative AI is strongest at bounded extraction, normalization, comparison and drafting. It can propose actor roles, asset fields, questions, flow steps and risk categories. Retrieval can help locate passages in approved documents. Code assistance can automate register checks and diagrams.

These capabilities do not establish truth. Models may invent mandates, merge institutions, overlook temporal qualifications or treat reported claims as facts. Their output is best understood as candidate work requiring source-level verification.

Capability	Appropriate role
Rules	Validate required fields and enforce explicit classifications.
Machine learning	Detect patterns or classify records when labelled evidence exists.
Generative AI	Draft structured candidates, summaries, code and alternatives.
Retrieval augmented generation	Answer within an approved source collection with citations.
Agentic workflow	Coordinate bounded tools with logging, permissions and stop conditions.
No AI	Use when evidence, risk or task structure does not justify assistance.

Prompt engineering as statistical specification

A prompt should specify purpose, context, evidence boundaries, output schema, uncertainty treatment and prohibitions. Vague prompts invite unsupported completion. Structured prompts make omissions visible and support repeatability across assets and reviewers.

Prompt quality is evaluated by the resulting workflow, not elegance of wording. Test prompts against representative, ambiguous and adversarial cases. Record model or tool version, parameters where relevant, source set, date, output and reviewer corrections. Reuse controlled patterns rather than copying unexamined text.

The statistician defines statistical meaning and acceptance criteria. Subject specialists define contextual conditions and consequences. AI may suggest prompt revisions, but it must not set the benchmark by which its own output is accepted.

Prompt component	Example requirement
Role and task	Extract candidate actors and roles from the supplied evidence only.
Evidence boundary	Use cited passages; do not use outside knowledge.
Schema	Return actor ID, role, source reference, uncertainty and question.
Uncertainty	Use Unknown when evidence is absent or conflicting.
Prohibitions	Do not infer legal authority or data quality from institutional names.
Verification	Flag every item requiring specialist confirmation.

What AI could do and what it should do

Capability does not imply appropriateness. AI could process public laws, strategy documents, metadata and interview notes rapidly. Whether it should do so depends on confidentiality, source quality, legal conditions, consequence, explainability, institutional capability and availability of independent review.

Task allocation should be granular. It may be appropriate to use AI to draft a candidate actor register from public documents but inappropriate to summarize restricted interview notes in the same system. It may draft a crosswalk while human experts approve equivalence. It may generate code while statisticians test logic and results.

Record the allocation decision before execution. A stopping rule identifies conditions requiring human takeover, a conventional method or termination. Examples include missing authoritative sources, high disagreement, protected data, unstable classifications or a consequence beyond the approved scope.

Field	Required statement
AI could do	Candidate capabilities under stated technical conditions.
AI should do	Approved tasks after proportionality, security and method review.
AI should not do	Excluded tasks and reasons.
Stopping rule	Observable condition that halts or redirects the workflow.
Human owner	Named role accountable for the decision and product.

The statistician and subject specialist retain authority

The statistician frames the problem, population, concepts, estimand, quality criteria, uncertainty and acceptance rules. The subject specialist confirms institutional mandates, domain meaning, operational practice, affected groups and plausible consequences. In some settings one person may hold both competencies, but the responsibilities remain distinct.

Human review is not a final cosmetic reading. It occurs before AI use through task design, during execution through challenge and correction, and after execution through independent verification and disposition. Reviewers must have authority and sufficient evidence to reject the output.

Responsibility cannot be delegated to the person who wrote the prompt if that person lacks the required statistical or domain competence. High-consequence decisions may require legal, confidentiality, cybersecurity or senior institutional approval in addition to statistical sign-off.

Responsibility	Non-delegable human decision
Statistical	Concepts, population, method, uncertainty, fitness and release.
Subject matter	Context, institutional reality, domain interpretation and consequences.
Data stewardship	Access, classification, metadata, retention and provenance.
Security and legal	Permitted environment, controls, compliance and escalation.
Leadership	Risk acceptance, resources, ownership and institutional action.

Independent verification and correction

Verification asks whether the output is supported, complete enough and fit for its purpose. It is independent when it relies on authoritative sources, reproducible tests, benchmarks or qualified review rather than asking the same model to confirm itself.

Use claim-level verification for mandates, numbers, definitions and dates. Use record-level sampling for large extracted registers. Use reconciliation for totals and identifiers. Use sensitivity analysis for thresholds and classification choices. Record detected failures and corrections because they reveal where future automation requires controls.

A corrected output is not automatically acceptable. Review whether the failure was isolated or systematic, whether the correction process is reproducible and whether similar errors could remain undetected. Residual limitations must follow the product into downstream use.

Verification method	Use
Source check	Confirm claim against exact authoritative passage or record.
Reproduction	Run code or transformation independently.
Benchmark	Compare against accepted reference cases or totals.
Dual review	Require separate statistical and subject-matter judgement.
Adversarial test	Test ambiguity, contradiction, missing data and edge conditions.
Disposition	Accept, revise, restrict, escalate or reject.

Failure modes automation bias and stopping rules

Common failures include fabricated mandates, incorrect role assignments, omitted actors, conflated assets, false equivalence, temporal confusion, confident completion of unknown fields, insecure data handling and code that runs but implements the wrong statistical rule.

Automation bias occurs when reviewers accept plausible output because it is fast, detailed or machine-produced. Anchoring occurs when the candidate structure narrows later investigation. Mitigate these risks by defining criteria before viewing output, using blind or independent checks for critical items and requiring reviewers to document disagreement.

Stopping rules protect both quality and time. Repeated unsupported claims, missing authoritative evidence, high correction rates, privacy uncertainty or inability to reproduce results are reasons to halt AI use. Continuing manually may be more efficient than repairing unreliable automation.

Failure signal	Required response
Unsupported authority claim	Reject claim and check the governing instrument.
Conflicting source interpretation	Escalate to qualified specialist and preserve both readings.
High extraction error rate	Stop batch processing and revise or abandon the method.
Protected information detected	Stop, contain and follow incident procedures.
Unreproducible output	Restrict use and return to controlled workflow.
No quality-adjusted gain	Use the conventional method.

Measuring quality adjusted efficiency

Efficiency is not raw speed. Measure total effort required to obtain an accepted product, including preparation, prompting, verification, correction, escalation and rework. Compare against a credible baseline performed at an appropriate level of competence.

Quality must be evaluated on criteria established in advance: accuracy, completeness, traceability, reproducibility, timeliness and fitness for use. A faster draft that requires extensive specialist repair may yield no gain. Conversely, AI may improve completeness and documentation even when elapsed time changes little.

Report variation across task types. Extraction from standardized public documents may yield consistent gains, while legal interpretation or ambiguous institutional mapping may not. Avoid universal productivity claims from a single exercise.

Measure	Definition
Baseline effort	Time and resources for the conventional workflow.
AI preparation	Source preparation, configuration and prompt design.
Execution	Tool run time and human interaction.
Verification	Source checks, tests and specialist review.
Correction and rework	Effort to repair failures and repeat affected steps.
Accepted quality	Result against predefined criteria and disposition.
Net conclusion	Faster, better, neutral or worse under stated conditions.

Workflow 1 AI assisted actor and authority mapping

Begin conventionally by defining the scope, collecting authoritative instruments and creating the actor schema. Then use an approved AI capability to extract candidate actors, roles, mandates and citations. Require the model to preserve uncertainty and avoid inferring authority from institution names.

The statistician checks whether all necessary roles are represented and whether the mapping supports the statistical product. Institutional and legal specialists verify mandates, current delegations and operational practice. Conflicts become explicit questions rather than model-resolved answers.

Compare the AI-assisted register with a manually produced sample. Record omission, commission, wrong-role and wrong-citation errors. Accept automation only for bounded extraction if correction rates and consequences remain acceptable.

Stage	Evidence
Baseline	Manual extraction from a representative source subset.
Prompt	Task, source boundary, schema, uncertainty and prohibition record.
Candidate	Actor-role records with exact citations.
Verification	Mandate and current-practice checks by authorized specialists.
Correction	Error log and revised records.
Efficiency	Quality-adjusted comparison and reuse decision.

Workflow 2 AI assisted asset inventory

Define the asset record and population of documents before extraction. AI can identify candidate assets, owners, concepts, populations, periods, formats and dependencies from catalogues and technical documentation. It can normalize labels and flag missing fields.

The principal risk is false equivalence. Similar titles may refer to different waves, populations or outputs. A publication may be mistaken for the underlying dataset. The statistician verifies asset identity, concepts and fitness; source owners confirm operational provenance and access conditions.

Use record sampling and reconciliation. Compare counts with catalogues, inspect all high-consequence fields and retain Unknown where evidence is absent. Do not allow the model to manufacture completeness.

AI contribution	Human control
Extract candidate fields	Approve schema and authoritative source set.
Normalize names	Confirm identity and version.
Suggest type classification	Validate source, product and reference-asset distinctions.
Flag omissions	Investigate with source owners.
Draft summaries	Approve fitness conclusion and permitted use.

Workflow 3 AI assisted flow and lineage mapping

Collect process descriptions, code, schedules, interface specifications and staff accounts. Define the required flow elements. AI can draft candidate steps, transformations and handoffs and can compare documents for inconsistent sequences.

Actual operations may diverge from documentation. Operators and statisticians walk through representative cases, including late data, corrections and revisions. Technology staff verify interfaces and logs. Each step receives actor and asset identifiers, control ownership and a fallback.

Use AI-generated diagrams only after the structured flow table is verified. The diagram is a view of controlled records, not the source of truth. When uncertainty remains, show it explicitly rather than smoothing the workflow.

Check	Question
Completeness	Are trigger, source, transformation, control, timing and destination recorded?
Reality	Does the documented process match observed practice?
Authority	Who accepts errors and approves progression?
Lineage	Can downstream products be identified after a change?
Resilience	Is the fallback tested and resourced?
Security	Are sensitive interfaces and details appropriately restricted?

Workflow 4 user need and fitness matrix

Translate each user decision into explicit requirements before examining AI opportunities. AI can organize consultation notes, propose quality dimensions and identify apparent mismatches between needs and assets. It cannot decide acceptable error or consequence on behalf of users.

Users, statisticians and domain specialists review the matrix. Separate essential thresholds from preferences. Record conflicts among users and distributional consequences. A national aggregate suitable for planning may be unfit for district targeting.

Verification compares every requirement with evidence from asset and flow records. A missing quality report is not evidence of poor quality, but it is an assurance gap. The disposition should distinguish fit, conditionally fit, not fit and unknown.

User requirement	Asset evidence	Disposition
Decision and population	Concepts, coverage and unit.	Fit only when population matches.
Timing	Production schedule and observed delays.	Conditional when lateness affects action.
Accuracy	Design, benchmark, error and revision evidence.	Unknown when evidence is absent.
Interpretability	Metadata, methods and user guidance.	Not fit when meaning cannot be defended.
Access	Release, confidentiality and technical route.	Conditional on lawful controlled access.

Workflow 5 screening AI opportunities

An ecosystem map supports opportunity selection by exposing prerequisites. Start with the decision and current workflow. Define the baseline problem, affected users, available evidence and responsible authority. Then consider alternative interventions, including non-AI process changes.

AI is justified when the task is sufficiently bounded, evidence and verification are available, risks are controllable and the expected gain exceeds lifecycle costs. A pilot should test a specific claim under predefined criteria rather than demonstrate that a tool can produce output.

The decision may be Proceed to scoping, Investigate dependencies first, Use a non-AI improvement or Do not pursue. Conditions, owners, stopping rules and fallback must accompany the recommendation.

Screen	Question
Purpose	What exact decision and statistical product improve?
Baseline	What current time, cost, error and control evidence exists?
Data	Are sources lawful, representative, documented and stable?
Quality	Which benchmarks, domains and tolerances apply?
Authority	Who owns method, review, release and consequences?
Sustainability	Are skills, funding, infrastructure and exit options credible?

Republic of Kintaba case context

Kintaba is a fictional lower-middle-income country of 24 million people. Agriculture employs a large rural population and is exposed to drought, pests and price volatility. Government requires district crop-production estimates six weeks earlier for food-security planning. The case contains deliberate gaps and contradictions.

The national statistics directorate runs an agricultural census every ten years and an annual production survey based on the last census frame. The ministry agriculture statistics unit publishes seasonal estimates from extension reports and a farmer sample, but methods differ by region. Extension coverage is better near roads.

Other actors include land administration, meteorology, the food-security secretariat, research institutes, a university geospatial laboratory, market information services, a mobile network operator, farmer organizations and the government digital service. No single actor controls the entire chain.

Decision pressure	Known constraint
Earlier district estimates	Annual survey release is too late for alerts.
Small-domain detail	Frames and source coverage may omit remote and irrigated holdings.
Source integration	No common identifier links holdings, parcels, extension reports and survey units.
Sustainability	Geospatial work has limited funding and specialist capacity.
Trust	Competing maize estimates lack visible definitions and revision status.

Kintaba evidence register

Treat each statement according to its evidence status. The following records are sufficient for the laboratory but do not resolve every authority, quality or access question. Learners must not fill gaps with assumed national practice.

EVD-01 assigns national coordination to the statistics directorate and permits designated sector producers. EVD-02 requires the ministry unit to transmit microdata and metadata within ninety days, but only tables were transmitted last year. EVD-03 records conflicting maize estimates of 4.8 and 4.2 million tonnes without definitions or revision status.

EVD-04 reports 86 percent crop-classification accuracy on 600 labelled fields in three accessible districts. EVD-05 reports higher trial yields under controlled water management without external-validity assessment. EVD-06 records the six-week timing gap. EVD-07 records the absence of a common identifier. EVD-08 records eight months of geospatial funding and one trained analyst.

Evidence problem	Required response
Conflicting estimates	Trace definitions, methods, authority and revision status.
Accuracy headline	Request class, district, crop and season performance plus reference-label design.
Trial result	Separate experimental efficacy from population effectiveness.
Missing identifier	Assess linkage alternatives, error and governance.
Temporary capability	Include continuity, transfer, documentation and fallback in the decision.

Practical 1 actor authority and evidence map

Time: 75 minutes. Produce an actor register and authority analysis for the Kintaba crop-estimation problem. First complete a conventional extraction for ACT-01 through ACT-04. Then use an approved AI tool on the remaining case evidence with the structured prompt below.

Prompt pattern: Using only the supplied case, return actor ID, candidate roles, supporting evidence ID, evidence label, unresolved authority question and confidence rationale. Use Unknown where authority is not established. Do not infer legal powers from names. Flag every record requiring specialist confirmation.

Verify every role assignment. Identify at least three authority questions that must be resolved before integration and one missing actor. Compare the manual and AI-assisted portions for omissions, wrong roles, unsupported inferences and review time.

Required submission	Evidence
Actor register	Multiple roles permitted; each assignment labelled E, R, U or I.
Authority analysis	Decision rights for concepts, access, method, release and escalation.
Prompt record	Exact prompt, tool context, output and revisions.
Verification log	Source checks, errors, corrections and unresolved questions.
Efficiency note	Baseline and AI-assisted quality-adjusted effort.

Practical 2 asset inventory and source validation

Time: 90 minutes. Identify at least twelve assets across surveys, administrative records, research, geospatial, private, reference and statistical-product categories. Distinguish institutions, datasets, metadata, models and publications.

Perform the first four records manually. Use AI to draft candidate records for the remainder from the Kintaba evidence. Require stable asset IDs, type, owner, population or coverage, time, identifiers, transformation, principal quality issue, access condition and verification question.

Challenge the candidate inventory. Identify a timely but potentially biased source, an experimental result generalized beyond evidence, a missing reference asset and two sources that cannot yet be integrated. Subject specialists must confirm operational meaning; the statistician approves fitness conclusions.

Asset decision	Meaning
Fit	Evidence supports the stated use under current conditions.
Conditionally fit	Use is permitted only with explicit limitations or controls.
Not fit	Known mismatch or risk prevents the stated use.
Unknown	Evidence is insufficient; identify the owner and required check.

Practical 3 critical flow and failure analysis

Time: 90 minutes. Trace the flow from field conditions to a cabinet food-security decision. Include at least five actors, three assets and six transformations or handoffs. Record controls, timing, failure modes, responsibility and fallback.

Create a manual high-level flow, then ask AI to challenge it for missing handoffs, implicit transformations and failure scenarios. The AI output is a test generator, not evidence. Verify suggestions against case records and mark unsupported items as hypotheses.

Identify the two most consequential failure points. For each, state the affected decision, consequence, present evidence, missing evidence, responsible human role, non-AI improvement and any bounded AI assistance. Explain how uncertainty propagates downstream.

Step field	Required content
Source	Actor and asset IDs.
Transformation	Editing, coding, linkage, modelling, aggregation or interpretation.
Control	Check, approval, access rule and owner.
Timing	Deadline, delay and synchronization.
Failure	Bias, loss, duplication, inconsistency, delay or unauthorized access.
Fallback	Alternative source or procedure and activation authority.

Practical 4 AI opportunity and pilot decision

Time: 90 minutes. Evaluate the proposal: Use generative AI and satellite imagery to produce district crop estimates automatically. The wording intentionally combines distinct technologies and transfers excessive authority to automation.

Reframe the proposal into candidate tasks such as metadata extraction, image classification, anomaly detection, documentation or narrative drafting. For each, state what AI could do, what it should do, what the statistician must do and what subject specialists must do. Identify verification evidence and stopping rules.

Produce one decision: Proceed to bounded scoping, Investigate dependencies first, Prefer non-AI improvement or Do not pursue. Define the pilot claim, baseline, domains, benchmarks, tolerance, human oversight, information protection, sustainability and fallback.

Pilot element	Required specification
Claim	Specific performance or efficiency improvement to test.
Baseline	Current method and accepted quality evidence.
Evaluation	Domains, metrics, tolerances and uncertainty.
Oversight	Named statistician and specialist review points.
Critical failure	Conditions that stop the pilot or prohibit release.
Decision pack	Evidence, limitations, recommendation and accountable sign-off.

Notebook and laboratory alignment

The accompanying notebook operationalizes the lesson; it does not replace it. Its data are synthetic and represent the Kintaba case. Cells should separate source records, conventional analysis, AI-assisted steps, verification, correction and final products.

Every AI interaction must be reproducible to the extent permitted by the tool. Record prompt, context, date, capability or model, configuration, output and reviewer disposition. When a live AI service is unavailable, learners can work with controlled candidate outputs and perform the same verification tasks.

The notebook should produce structured exports for the actor register, asset register, flow table, needs matrix, risk register and opportunity decision. Automated validation checks required fields and identifiers. Statistical and subject-matter acceptance remains a documented human action.

Notebook section	Output
Setup and data classification	Approved-use decision and synthetic data confirmation.
Baseline workflow	Manual sample and recorded effort.
AI-assisted workflow	Prompted candidate records and timing.
Verification	Tests, source checks and error log.
Professional product	Validated registers and ecosystem sketch.
Reflection	Quality-adjusted comparison and transfer conditions.

Assessment strategy and rubric

Assessment measures the complete performance cycle. Learners submit the ecosystem product, evidence trace, prompt record, verification log, corrections, efficiency comparison and professional recommendation. An oral defence tests whether they understand the evidence and can reject inappropriate automation.

Scoring should not reward fluent presentation when source traceability or statistical validity is weak. Critical failures override compensatory scores. These include fabricated evidence, unapproved disclosure, AI as sole verifier, absent human accountability and release of an unfit product.

A competent submission can contain unresolved unknowns when they are correctly identified, consequentially assessed and assigned to an evidence owner. False certainty is a more serious deficiency than an explicit evidence gap.

Criterion	Weight	Minimum standard
Scope and statistical purpose	15	Decision, population, product and boundary are defensible.
Actors assets and flows	20	Records are complete enough, stable and source-traceable.
Quality risk and fitness	20	Error mechanisms and consequences support the conclusion.
AI task allocation and prompting	15	Use is bounded, reproducible and appropriate.
Verification and correction	20	Independent checks identify and resolve material failures.
Judgement and communication	10	Recommendation, limits and accountability are clear.

Workplace assignment and specialist review

Apply the method to an approved national domain or product. Limit the first assignment to a boundary that can be evidenced within four to six weeks. Examples include agricultural early warning, business registers, health service statistics or education management information.

Begin with an authorization note stating purpose, data classification, approved tools, reviewers and exclusions. Use public or controlled institutional evidence. Interview records should distinguish respondent statements from verified facts. Do not place sensitive operational material in the learner submission unless explicitly authorized.

The final review includes a lead statistician and at least one relevant subject specialist. Add data stewardship, legal or security review when conditions require. The institution decides whether the product is accepted, restricted, revised or used to scope a later diagnostic or pilot.

Gate	Evidence
Authorization	Scope, owner, reviewers, data class and tool conditions.
Design	Record schemas, baseline workflow and acceptance criteria.
Execution	Evidence register, prompts, outputs, verification and corrections.
Product	Map, registers, flow, fitness matrix and opportunity decision.
Sign-off	Named reviewers, residual limitations and disposition.

Implementation variants and localization

Large institutions may distribute roles across methodology, domain, data stewardship, legal, security and technology teams. Smaller institutions may combine roles and use pooled regional expertise. Separation of final approval should be maintained where feasible, with external review for high-consequence work when internal independence is limited.

Localization may change laws, institutions, classifications, examples, languages and approved tools. It must not weaken the core outcome, evidence labels, quality criteria, non-substitution principle or assessment of observable performance. Translation should be reviewed by a subject specialist and a language professional familiar with official statistics.

Low-resource delivery may use offline documents and controlled AI outputs rather than live services. Learners still practise task allocation, prompting, verification and correction. Tool unavailability is not a reason to omit AI competence; it changes the laboratory implementation.

Context	Adaptation
Advanced system	Automated extraction, controlled retrieval, model monitoring and independent evaluation.
Medium capability	Approved general tools with redaction, structured review and limited automation.
Small institution	Combined roles, minimum viable laboratory and external specialist review.
Low connectivity	Offline source pack, recorded outputs and reproducible verification.
Multilingual	Controlled translation, terminology governance and bilingual specialist review.

Synthesis and professional checklist

A defensible ecosystem map connects public purpose to actors, assets, flows, quality, safeguards, capability and risk. It makes authority and uncertainty visible. It supports action because it distinguishes structural dependencies from problems that a bounded technical intervention can address.

AI improves the workflow when it accelerates structured extraction, comparison, test generation, documentation or code without weakening evidence and accountability. Its contribution is measured after verification and correction. The human contribution is not residual; it defines meaning, acceptable evidence, consequence and disposition.

Before completing the module, confirm that the professional product meets every item below and that the accountable reviewers can locate the evidence.

- The decision, population, geography, time and exclusions define the boundary.
- Actors and roles are separate, source-traceable and current enough for the use.
- Assets include provenance, concepts, coverage, identifiers, quality, access and versions.
- Flows record transformations, controls, delays, failures, responsibility and fallback.
- User needs are translated into explicit fitness-for-use requirements.
- AI use has a baseline, structured prompt or configuration, verification, correction and stopping rules.
- Statistician and subject-specialist duties are explicit and non-delegable.
- Quality-adjusted efficiency and residual limitations are documented.

Glossary and authoritative references

Key terms. Ecosystem: the connected institutional, informational, technical and governance arrangements through which information is produced and used. Asset: a governed body of data, metadata, code, documentation or output. Flow: a traceable sequence of sources, transformations, controls and handoffs. Fitness for use: adequacy for a stated decision under stated conditions. Verification: independent evidence or testing that supports acceptance. Disposition: accept, revise, restrict, escalate or reject.

Authoritative reference families for further study include the United Nations Fundamental Principles of Official Statistics; the United Nations National Quality Assurance Frameworks Manual for Official Statistics; the UNECE Generic Statistical Business Process Model; statistical confidentiality and data stewardship guidance; domain statistical standards; and applicable national law, classifications and institutional policies.

For this pathway, the controlling curriculum documents are the Phase 2A Curriculum Blueprint and Competency Architecture, the SRAI Official Statistics and AI Pathway Production Reference version 1.3, and the Pathway-Wide AI-Assisted Statistical Work Standard version 1.0. Current institutional sources always govern local mandates, access and release authority.

Completion record	Requirement
Learner	Name, cohort and submission identifiers.
Product	Controlled ecosystem map and supporting registers.
Reviewers	Statistician, subject specialist and other required authorities.
Decision	Accepted, revision required, restricted or rejected.
Limitations	Unresolved evidence and conditions on use.
Next action	Named owner, evidence required and review date.